

VIVEK

Phone: 9870710209 | Email: vivekrajraghav@gmail.com, vivek1@mfs.iitr.ac.in

[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Area of Interest: Machine Learning, Deep Learning, Predictive Modeling, Data Structures & Algorithms

EDUCATION

Degree/Examination	Institution/Board	Year	CGPA/%
M.Tech in Data Science (I Year I Sem)	Indian Institute of Technology Roorkee	2028	-
B.Tech in Biotechnology	Sardar Vallabhbhai Patel University of Agriculture and Technology, Meerut	2026	8.250
Intermediate (Class XII)	Alpine Public School, Khurja (CBSE Board)	2021	92.00%
Matriculate (Class X)	D.S. Memorial Oxford Public School, Shikarpur (CBSE Board)	2019	82.16%

SKILLS

Programming & Scripting: Python, Linux/WSL, AWS CLI

Data & AI Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit Learn, Plotly, LangChain, ChromaDB, Ollama

Tools & Pipelines: Streamlit, Snakemake, Groq API, Pydantic, Conda, uv

Bioinformatics & Domain: Fastp, Megahit, Kraken2, MAFFT, FastTree, WGS, 16S rRNA, Immunoassays, Cell Culture

EXPERIENCE

Research Internship | INMAS DRDO, New Delhi

February 2026 - May 2026

Topic: Effect of LDI-2000 Irradiation on Gut Microbiota of C57BL/6 Mice

- Analyzed 16S rRNA metagenomic data to quantify radiation-induced gut microbiome shifts and taxonomic population dynamics in murine models.
- Executed comprehensive wet-lab workflows encompassing bacterial phenotypic characterization, bacteriophage isolation, and kinetic assays to evaluate complex phage-host interactions.

PROJECTS

Microbiome Analytics Platform with RAG Assistant | Self Project

June 2026 - Present

- Engineered an automated bioinformatics pipeline using Snakemake to process WGS and 16S rRNA sequencing data, integrating Fastp, Megahit, Kraken2, MAFFT, and FastTree.
- Developed a local RAG AI assistant leveraging ChromaDB, LangChain, and Ollama (Llama 3.2, Nomic-Embed-Text) for semantic querying of taxonomic profiling and quality control metrics.
- Built an interactive frontend using Streamlit, rendering dynamic relative abundance visualizations and phylogeny mapping with Plotly, Pandas, and NumPy.

- Configured high-concurrency AWS CLI fetching to source memory-intensive Kraken databases and containerized the workflow using Conda in a Linux/WSL environment.

AI-Powered Resume Analyzer & Scorer | Self Project2026

- Developed an automated, LLM-driven recruitment tool using Python and the Groq API (openai/gpt-oss-120b) to extract structured data from Job Descriptions and candidate resumes.
- Engineered a dual-prompting scoring system to evaluate candidate skills, calculate a 0-100% match score, and generate ranked outputs with detailed AI verdicts.
- Enforced structured JSON data extraction using Pydantic schemas to eliminate LLM hallucinations.

LICENSES & CERTIFICATIONS

- **Monitoring in Google Cloud Skill Badge** – Google (Issued Aug 2026)
- **Analyze Sentiment with Natural Language API Skill Badge** – Google (Issued Aug 2026)
- **Immunology** – NPTEL, IIT Kharagpur (Issued Oct 2024) | Credential ID: NPTEL24BT63S655600650
- **Bioreactors** – NPTEL, IIT Madras (Issued Aug 2024) | Credential ID: NPTEL24BT39S141600077
- **Human Molecular Genetics** – NPTEL, IIT Kanpur (Issued Apr 2023) | Credential ID: NPTEL23BT10S45160686

POSITIONS OF RESPONSIBILITY & EXTRA CURRICULARS

NSS Volunteer | SVPUA&T, MeerutAugust 2022 - July 2024

- Dedicated National Service Scheme (NSS) Volunteer at SVPUA&T Meerut with 240 hours of formal community service.
- Actively engaged in organizing social welfare initiatives and participated in special intensive camps focused on rural outreach and local development.

REFERENCES

Dr. Sonika Sharma, Scientist E | INMAS, DRDO, New Delhi
Email: ssharma.inmas@gov.in | Phone: +91 8811014929